Readiness to Train

Prescriptive Analytics for Optimal Training Intensity

KU Leuven & OH Leuven
Project Overview & Problem Statement

1. Core Objective

The goal of this project is to **optimise the performance of players on match day** by developing a prescriptive analytics system that estimates individual player **Readiness to Train**. Using **Causal Machine Learning** within a **Dynamic Treatment Regime (DTR)** framework, the system will recommend an optimal daily training intensity for each player — delivered as a continuous score between 0 and 1. A score of 0 indicates the player should rest; a score of 1 indicates the player can train at maximal intensity.

The system will be fully **individualised**: each player will receive a personalised score derived from their own physiological profile, training history, and evolving daily state. The ultimate objective is to find the training intensity sequence across the pre-match cycle that maximises each player's physical output on match day.

2. Partnership & Data

The project is a collaboration between **KU Leuven** (methodology, analysis) and **OH Leuven** (professional football club, data provider). OH Leuven operates a systematic daily player monitoring programme covering GPS load metrics, subjective wellness questionnaires, medical status, and match performance. Four raw datasets are available:

Dataset	Rows	Cols	Players	Start	End	Days
Readiness_Data	14,359	24	28	2024-07-01	2026-02-17	597
Raw_Data	9,968	39	84	2024-05-02	2026-03-01	670
Sessions	1,206	9	—	2024-05-02	2026-03-01	670
Games	403	9	24	2025-07-27	2026-02-28	216

The **processed dataset** (RTT.xlsx) merges all four sources into a single analysis-ready file with 46 engineered features per player-day. Readiness_Data is the base; Raw_Data provides detailed GPS/HR at session level (shifted +1 day so it appears as "yesterday" data); Games provides match performance data (also shifted +1 day). Player overlap: all 28 Readiness_Data players appear in Raw_Data; 23 of 28 appear in Games.

Data dictionaries: Two auto-generated PDF documents describe every variable in detail. The **Raw Data Dictionary** (*data/raw/Raw Data Dictionary.pdf*) documents all columns across the four raw datasets. The **RTT Data Dictionary** (*data/processed/RTT Data Dictionary.pdf*) documents all 46 columns in the processed dataset, including engineered features and their temporal positions.

Temporal ordering within each row: Variables in a single row have different temporal positions. "Yesterday" columns contain data from day t–1 (fully known). Morning wellness z-scores are measured at the start of day t (before any activity decisions). Post-assessment variables (Activity Type Today, Selected) are determined after the morning assessment and represent the treatment decision — they must never be used as predictors in standard ML.

3. Sports Science Foundation

Training stress produces fatigue, which during recovery is followed by adaptation (supercompensation). The ACWR (Acute:Chronic Workload Ratio) captures where a player sits on this curve: the sweet spot is 0.8–1.3; values above 1.5 are flagged as danger zones (Gabbett, 2016). GPS metrics are individualised as percentages of each player's match benchmarks. The training intensity score A(t) is the tanh-compressed harmonic mean of these GPS percentages, soft-capped in [0, 1). The typical weekly structure (5–6 training days followed by a match) creates player-specific match cycles central to the DTR formulation.

4. Why This Is a Causal Problem

Standard predictive modelling asks *"given the data, what will happen?"*. This project asks *"given a player's full history up to today, what training intensity today best contributes to future match-day performance?"*. This requires causal reasoning for three reasons:

1. Time-Varying Confounding Affected by Prior Treatment. Covariates at time t (fatigue, readiness, ACWR) are simultaneously *consequences* of treatment at t–1 and *causes* of treatment at t. Classical regression cannot handle this correctly — this is precisely the setting for which G-methods were developed.

2. Treatment-Confounder Feedback. Coaching decisions are observational. Hard sessions are prescribed when players appear fresh; recovery is prescribed when fatigue is high. The observed training-performance association conflates physiological response with coach selection behaviour.

3. Sequential Decision-Making. The treatment is a repeated, time-varying intervention. The question is not just "what training today?" but "what *sequence* of training intensities over the cycle maximises match-day performance?" — a DTR problem.

5. Causal Framework

Symbol	Role	Description
L(t)	Covariates	Player state at time t: wellness, ACWR, GPS load, schedule position
A(t)	Treatment	Training intensity [0,1]: tanh(harmonic mean of GPS benchmark %)
Y	Outcome	Match-day performance: high-intensity metrics per ball-in-play minute

The problem is modelled as a **longitudinal causal DAG** (Directed Acyclic Graph) with the following structure within each match cycle:

- At each time step t, the player's **covariates L(t)** (morning wellness, fatigue, ACWR) causally affect the coaching staff's **treatment decision A(t)** (what training intensity is prescribed that afternoon).
- The treatment A(t) then causally affects the player's **state L(t+1)** the next morning (training changes fatigue, soreness, adaptation level).
- This creates a **feedback loop**: L(t) → A(t) → L(t+1) → A(t+1) → ... repeating across the match cycle until match day, when the final player state and cumulative training determine the **match outcome Y**.
- Between cycles, the match outcome Y feeds back into the first state of the next cycle (match fatigue carries over).

The figure below shows an example DAG for Player 1, match cycle 4 (schematic view with grouped covariates):

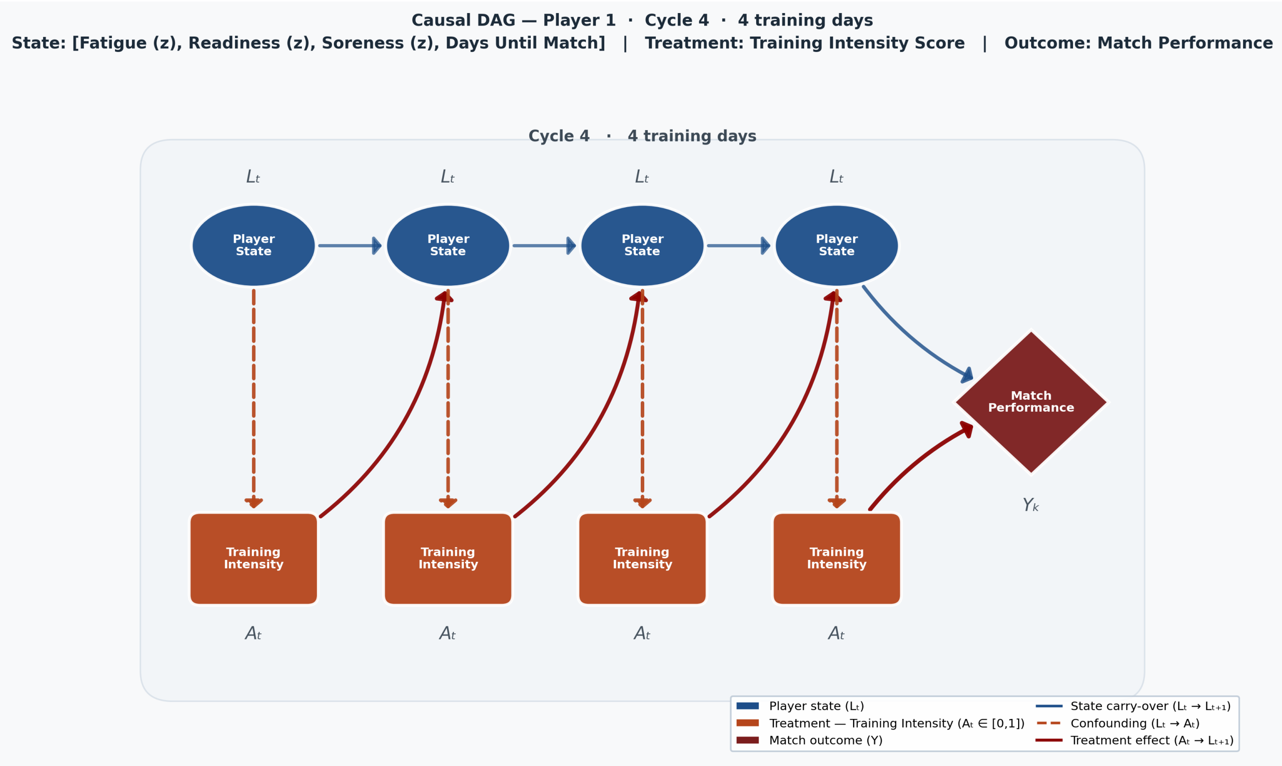

Figure: Causal DAG for Player 1, match cycle 4 (schematic). Blue ellipses = player state L(t), amber rectangles = treatment A(t), crimson diamond = match outcome Y.

This DAG structure is generated automatically for each player and each match cycle using the DAGCreator class (499 DAGs total, stored in *images/DAGs/player X/*). It encodes the causal assumptions needed for G-methods: confounding edges (L → A), treatment effects (A → L at t+1), state carryover (L → L at t+1), and inter-cycle feedback (Y → L at cycle k+1).

6. Key Challenges

Small N, Large T

Only 28 players are monitored in Readiness_Data, but each player has up to 597 days of observations. The signal is likely dominated by **within-player dynamics**. Models must be parsimonious or leverage partial pooling to avoid overfitting across players. Methods must balance individual-level tailoring with the limited between-player sample size.

Time-Varying Confounding Affected by Prior Treatment

Covariates L(t) at time t are simultaneously consequences of treatment at t–1 and causes of treatment at t. This is the defining feature of the problem and the reason standard regression fails. The appropriate causal identification framework is the theory of G-methods (Robins, 1986): G-computation, Marginal Structural Models with IPTW, or G-Estimation / Structural Nested Mean Models.

Observational Data with Coach Selection Bias

Training loads are prescribed by coaching staff based on observed player state. Players who appear fresh receive harder sessions; fatigued players receive recovery sessions. This creates a systematic association between baseline health and training dose that confounds any naive regression of load on performance.

Sparse Match-Day Outcomes

The outcome Y (match performance) is observed only on match days. With a 1-week cycle, each player contributes approximately one outcome observation per 6 rows of covariate/treatment data. The Games dataset covers 24 players over 27 match dates (403 player-match rows). Mean minutes played: 67.4 min/match; mean high-intensity distance per BIP: 18.75 m/min.